

Regression Test Log

ISSUE-D7-01 — Traffic Light Minimum Duration

Purpose

Confirm the load-vs-decrement priority logic correctly handles a state duration of 1 cycle — the minimum realistic value — without skipping the state or hanging for an extra cycle.

Test

`tb/traffic_light_min_duration_tb.sv` instantiates `traffic_light` with `MAIN_YELLOW_TIME(1)` overridden (other durations left at default: 4, 6, 2). One clean pass through the 1-cycle phase fully answers the question; the run continues 16 cycles into the following phases for extra confidence.

Logging uses the corrected honest per-cycle scheme ([ISSUE-D5-03](#)) and exposes the internal `count`, so durations are read numerically, not by counting rows.

Result

Verbatim — corrected RTL + corrected testbench.

```
VCD info: dumpfile dump.vcd opened for output.
time  cycle  state  main_r  main_y  main_g  side_r  side_y  side_g  count
25000  1      MAIN_GREEN  0      0      1      1      0      0      3
35000  2      MAIN_GREEN  0      0      1      1      0      0      2
45000  3      MAIN_GREEN  0      0      1      1      0      0      1
55000  4      MAIN_GREEN  0      0      1      1      0      0      0
65000  5      MAIN_YELLOW 0      1      0      1      0      0      0
75000  6      SIDE_GREEN  1      0      0      0      0      1      5
85000  7      SIDE_GREEN  1      0      0      0      0      1      4
95000  8      SIDE_GREEN  1      0      0      0      0      1      3
105000 9      SIDE_GREEN  1      0      0      0      0      1      2
115000 10     SIDE_GREEN  1      0      0      0      0      1      1
125000 11     SIDE_GREEN  1      0      0      0      0      1      0
135000 12     SIDE_YELLOW 1      0      0      0      1      0      1
145000 13     SIDE_YELLOW 1      0      0      0      1      0      0
155000 14     MAIN_GREEN  0      0      1      1      0      0      3
165000 15     MAIN_GREEN  0      0      1      1      0      0      2
175000 16     MAIN_GREEN  0      0      1      1      0      0      1
Min-duration regression test complete.
Day-05_03_traffic_light_min_duration_tb.sv:58: $finish called at 175000 (1ps)
```

Shaded rows are the YELLOW phases. Cycle 5 — the overridden 1-cycle `MAIN_YELLOW` — is the row this test exists to check.

Verdict

`MAIN_YELLOW` occupies exactly **one** cycle (cycle 5, `count=0`) between a 4-cycle `MAIN_GREEN` (count 3→0) and a 6-cycle `SIDE_GREEN` (count 5→0): the FSM enters it, holds for exactly the overridden duration, and advances on the next edge. **No skip, no extra hang cycle.**

A duration-1 phase is loaded with `count=0` and is immediately due for reload the next cycle — the load-vs-decrement priority (`count == 0` check taking priority over plain decrement) generalizes correctly to the smallest possible duration.

Incidental confirmation: this run also shows the reset fix from [ISSUE-D5-02](#) — the first `MAIN_GREEN` is a true 4 cycles (count 3, 2, 1, 0), matching the corrected reset load of `MAIN_GREEN_TIME-1`.

ISSUE-D7-01 is resolved.